ETC3250/5250 Introduction to Machine Learning

Week 10: Model-based clustering and self-organising maps

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Overview

We will cover:

- Models of multimodality using Gaussian mixtures
- Fitting model-based clustering
- Diagnostics for the model fit
- Self-organising maps and dimension reduction

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Model-based clustering

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From algorithm to models

- Last week we saw two clustering algorithms - k-means and hierarchical
- These were recipes for clustering observations rather than models that explain the data
- Similar to how k-nearest neighbours was a recipe for supervised learning prediction…
- … and Logistic regression or LDA where models

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Remembering LDA

- In supervised classification $y_i \in \{1, \ldots, K\}$ and $x_i \in \mathbb{R}^d$
- LDA assumed the distribution of the features for each **known** class was multivariate normal with class specific mean μ_k and pooled covariance Σ
- i.e. for i with $y_i = k$ $x_i \sim f_k(\cdot; \mu_k, \Sigma)$ with
$$f_k(x; \mu_k, \Sigma) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\{-\frac{1}{2}(x - \mu_k)^\top \Sigma^{-1} (x - \mu_k)\}$$
- used Bayes rule to produce a classification probability $P(y_i = k)$
- Estimated μ_k and Σ via maximum likelihood (class wise column means and pooled covariances)
- QDA relaxed equal variance and estimated class specific covariance Σ_k

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Mixtures and Clusters

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Overview

Model-based clustering makes the same assumption about the distribution of the features, except it doesn't know the class labels, primarily

- Assumes the data in each **unknown** cluster is a sample from a Gaussian distribution
- Results in a Gaussian Mixture Model to account for the cluster labels being **unknown**
- Requires the assumption that clusters have an elliptical shape
- The shape is determined by the variance-covariance of the clusters
- A variety of models is available using different constraints on the variance-covariance

The model

Model assumes that the distribution of the observed data has density

$$f(x_i; \{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K) = \sum_{k=1}^K \pi_k f_k(x_i; \mu_k, \Sigma_k)$$

where f_k is usually a multivariate normal distribution.
The parameters are estimated by maximum likelihood, and choice between models is made using BIC.
$\pi_k \in (0, 1)$ with $\sum_{k=1}^K \pi_k = 1$ allows us to estimate the proportion of observations in each cluster and account for unbalanced clusters

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Parametrisation of the var-cov matrices (1/2)

The covariance matrix Σ_k has $d(d+1)/2$ parameters (i.e. for $d = 10$, has 55 parameters per cluster, for $d = 50$, has 1225 parameters per cluster)

Constraints applied on cluster variance-covariance:

$\Sigma_k = \lambda_k D_k A_k D_k^\top$

- volume ($\lambda_k \in \mathbb{R}$): size of the cluster, i.e. one cluster could be half the size of another
- shape ($A_k \in \mathrm{diag}(d)$): difference variances in each dimension
- orientation ($D_k \in \mathbb{R}^{d \times d}$): aligned with axes (low covariance) or not (high covariance)

- $\Sigma_k = \lambda I$ for all k is model 1, EII
- $\Sigma_k = \lambda D A D^\top$ for all k is model 7, EEE
- $\Sigma_k = \lambda D_k A D_k^\top$ is model 11, EEV

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Model	Family	Volume	Shape	Orientation	Identifier
1	Spherical	Equal	Equal	NA	EII
2	Spherical	Variable	Equal	NA	VII
3	Diagonal	Equal	Equal	Axes	EEI
4	Diagonal	Variable	Equal	Axes	VEI
5	Diagonal	Equal	Variable	Axes	EVI
6	Diagonal	Variable	Variable	Axes	VVI
7	General	Equal	Equal	Equal	EEE
8	General	Equal	Variable	Equal	EVE
9	General	Variable	Equal	Equal	VEE
10	General	Variable	Variable	Equal	VVE
11	General	Equal	Equal	Variable	EEV
12	General	Variable	Equal	Variable	VEV
13	General	Equal	Variable	Variable	EVV
14	General	Variable	Variable	Variable	VVV

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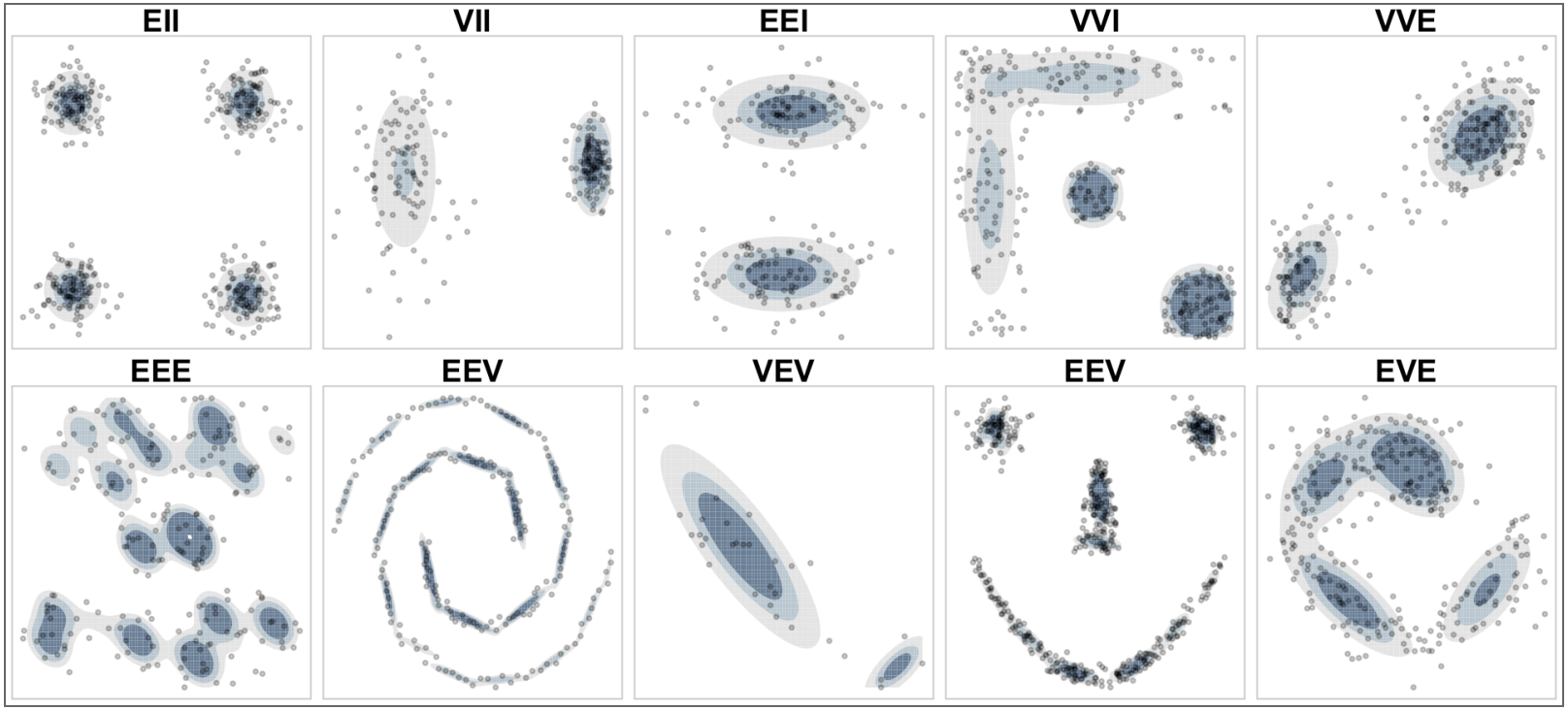
Parametrisation of the var-cov matrices (2/2)



Source: **Boehmke (2020) Hands-on machine learning**

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Reduction in the number of parameters to estimate

$\Sigma_k = \lambda_k D_k A_k D_k^\top$

- VVV estimates a full covariance matrix Σ_k for each cluster, the covariance matrix has $d(d+1)/2$ parameters and so we have $K \times d(d+1)/2$ parameters
- EII estimates $\Sigma_k = \lambda I$ for all k and so only has 1 parameter
- EEE estimates a full covariance matrix Σ but fixes this across clusters i.e. $\Sigma_k = \Sigma$ for all k so has $d(d+1)/2$ parameters

NOT EXAMENABLE

- EVE estimates $\Sigma_k = \lambda D A_k D^\top$ so we have
- 1 λ across all clusters
- 1 matrix D across all of the clusters, which to be a valid orientation/rotation has $d(d-1)/2$ elements and
- An A_k per cluster, A_k has $(d-1)$ elements as all of the other axes width are relative to the first axes width.

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- In total we have $1 + d(d-1)/2 + K(d-1)$ parameters

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Reduction in the number of parameters to estimate

$$f(x_i; \{\pi_k, \mu_k, \Sigma_k\}_{k=1}^{K}) = \sum_{k=1}^{K} \pi_k f_k(x_i; \mu_k, \Sigma_k)$$

In addition to Σ_k we must also estimate

- A $\mu_k \in \mathbb{R}^d$ for each cluster k (i.e. $K \times d$ parameters)
- Cluster weights $\pi_k \in \mathbb{R}^K$, while we have K clusters $\sum_{k=1}^{K} \pi_k = 1$ so $\pi_K = 1 - \sum_{k=1}^{K-1} \pi_k$ so there are only $K - 1$ parameters

You can't estimate a model with more parameters than observations!

Unless you regularise…

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Maximum likelihood

- Once a form/constraint for the covariance matrices has been decided (more on this later)
- We learn the parameters $\mu_k, \Sigma_k, k = 1, \ldots, K$ by maximising the likelihood
- Remember, the likelihood is the probability under the model of the observed data
- For the Gaussian mixture model

$$
\begin{aligned}
l_n(\{\mu_k, \Sigma_k\}_{k=1}^K) &= f(x_1, x_2, \ldots, x_n; \{\mu_k, \Sigma_k\}_{k=1}^K) \\
&= \prod_{i=1}^n f(x_i; \{\mu_k, \Sigma_k\}_{k=1}^K) \\
&= \prod_{i=1}^n \sum_{k=1}^K \pi_k f_k(x_i; \mu_k, \Sigma_k)
\end{aligned}
$$

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- The Maximum likelihood estimates are the parameter that made the observed data as likely as possible

$$\{\hat{\mu_k}, \hat{\Sigma_k}\}_{k=1}^{K} := \arg\max l_n(\{\mu_k, \Sigma_k\}_{k=1}^{K})$$

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Example: nuisance variable (1/4)

X_2 is a 'nuisance' variable that is not useful for clustering
Could cause a problem for k-means or hierarchical clustering

```
set.seed(712)
df_km2 <- kmeans(df, 2)
df_km <- df |>
  mutate(cl = factor(df_km2$cluster))

ggplot(data=df_km, aes(x1, x2, color = cl)) + geom_point(
    theme_bw() + theme(aspect.ratio=1)
```

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Example: nuisance variable (2/4)

X_2 is a 'nuisance' variable that is not useful for clustering

Could cause a problem for k-means or hierarchical clustering

Mclust

see also **vignettes/mclust**

```
df_mc <- Mclust(df, G = 2)
summary(df_mc)
```

```
----------------------------------------------------
Gaussian finite mixture model fitted by EM algorithm
----------------------------------------------------

Mclust EEI (diagonal, equal volume and shape) model with 2
components:

 log-likelihood   n df  BIC  ICL
           -204 100  7 -441 -441

Clustering table:
```

```
 1  2
50 50
```

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BIC

- The BIC stands for Bayesian Information Criteria
- For any variance/covariance constraints we use Maximum Likelihood to estimate the parameters $\hat{\mu_k}, \hat{\Sigma_k}$
- We can't use the likelihood to choose between models though, as the likelihood will always favour more complex models
- Information Criteria use the log-likelihood but also penalise model complexity

$$\text{BIC} = (\#\text{parameters})\log(n) - 2\log(l_n(\{\hat{\mu_k}, \hat{\Sigma_k}\}_{k=1}^K))$$

- $(\#\text{parameters})$ counts the number of separate parameter reqired for $\{\hat{\mu_k}, \hat{\Sigma_k}\}_{k=1}^K$
- $\{\hat{\mu_k}, \hat{\Sigma_k}\}_{k=1}^K$ are the maximum likelihood estimators
- n is the number of observations

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- l_n is the likelihood of observations for given parameters

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Example: nuisance variable (3/4)

```
plot(df_mc, what = "density")
```

Each of those circles is a set with the same density (i.e. equally likely within each cluster)

```
plot(df_mc, what = "uncertainty")
```

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Example: nuisance variable (4/4)

Cluster means

```
options(digits=2)
df_mc$parameters$mean
```

```
    [,1]  [,2]
x1 -0.97  0.97
x2  0.11 -0.11
```

Cluster variance-covariances

```
df_mc$parameters$variance$sigma
```

```
, , 1

      x1   x2
x1 0.052 0.00
x2 0.000 0.98

, , 2
```

```
x1 0.052 0.00
x2 0.000 0.98
```

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Example: nuisance cases (1/4)

The observations that ‘join’ the two clusters make hierarchical clustering hard

```
df_hc_w3 <- hclust(dist(df), method="single")

ggdendrogram(df_hc_w3, rotate = TRUE, size = 4)
```

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Example: nuisance cases (1/4)

The observations that 'join' the two clusters make hierarchical clustering hard

```
df_hc <- hclust(dist(df), method="single")

df_hc_w2 <- df |>
  mutate(cl = factor(cutree(df_hc, 2)))

ggplot(data=df_hc_w2, aes(x1, x2, color = cl)) + geom_poi
    theme_bw() + theme(aspect.ratio=1)
```

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Example: nuisance cases (2/4)

The observations that 'join' the two clusters make hierarchical clustering hard

```r
df_mc <- Mclust(df, G = 2)
summary(df_mc)
```

```
----------------------------------------------------
Gaussian finite mixture model fitted by EM algorithm
----------------------------------------------------

Mclust EEE (ellipsoidal, equal volume, shape and
orientation) model with 2 components:

 log-likelihood   n df  BIC  ICL
          -205 120  8 -447 -452

Clustering table:
```

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Example: nuisance cases (3/4)

```
plot(df_mc, what = "density")
```

```
plot(df_mc, what = "uncertainty")
```

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Example: nuisance cases (4/4)

Cluster means

```
df_mc$parameters$mean
```

```
    [,1] [,2]
x1 -0.88 0.92
x2 -0.88 0.92
```

Cluster variance-covariances

```
df_mc$parameters$variance$sigma
```

```
, , 1

      x1    x2
x1 0.186 0.081
x2 0.081 0.185

, , 2

      x1    x2
```

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```
x1 0.186 0.081
x2 0.081 0.185
```

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Example: penguins (1/2)

As well as selecting the covariace matrix constraints, BIC can be used to select the number of clusters

Let the model decide on the number of clusters, and parametrisation.

```
p_mc <- mclustBIC(p_std[,2:5], G=2:8)
summary(p_mc)
```

```
Best BIC values:
         VEE,4   VEE,5 VEE,3
BIC      -2497 -2503.7 -2510
BIC diff     0    -7.2   -14
```

Maybe there really is another sub-species of penguin…

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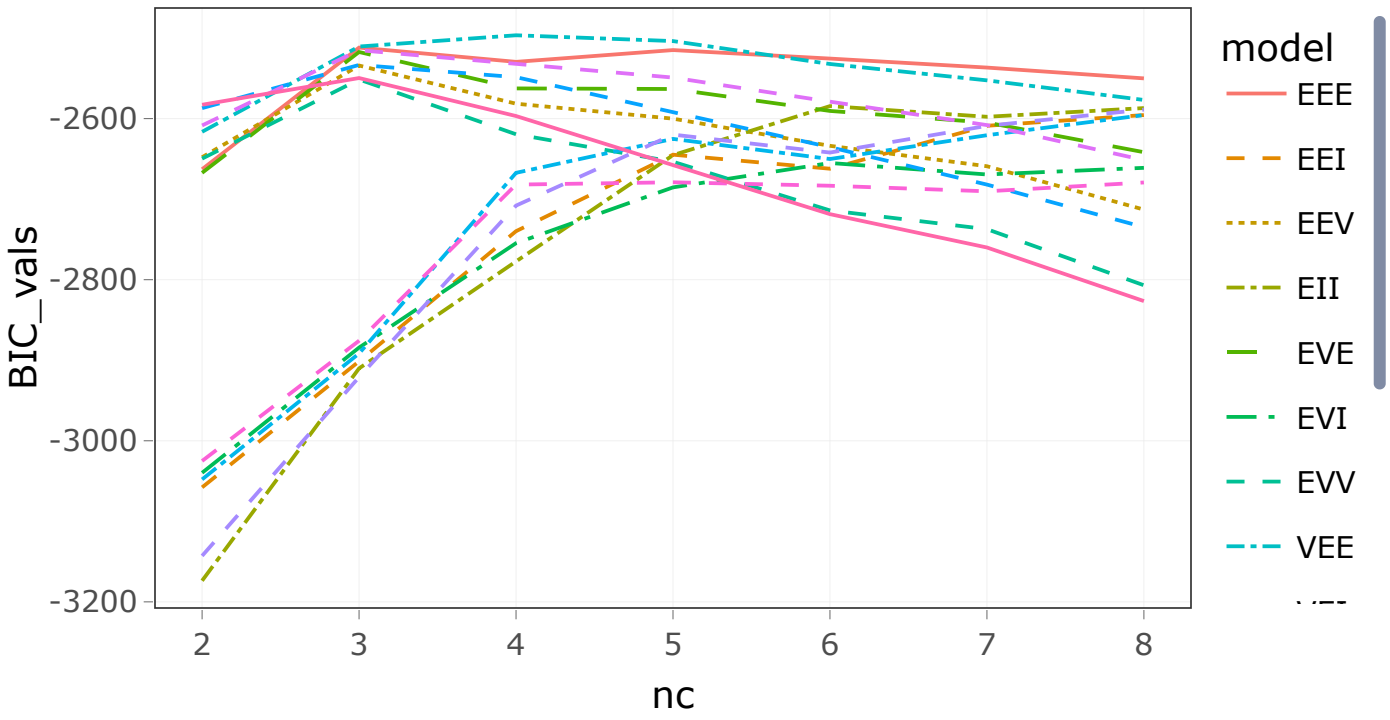
Example: penguins (1/2)

```
pmc1 <- ggmcbic(p_mc) +
  scale_x_continuous(breaks=1:7, labels=2:8)
ggplotly(pmc1)
```



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The model with the smallest BIC is the best. However, `mclust` negates the BIC so we look for the maximum.

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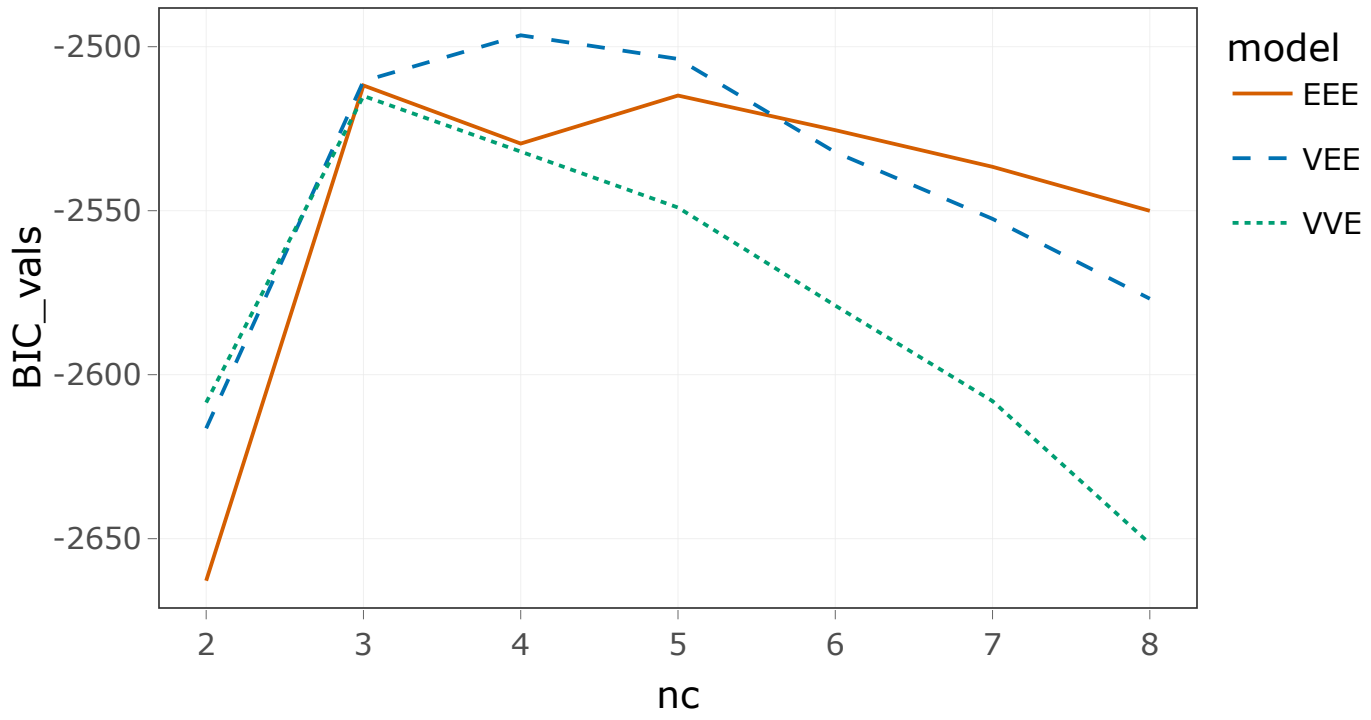
Example: penguins (1/2)

```
pmc2 <- ggmcbic(p_mc, top=6) +
  scale_x_continuous(breaks=1:7, labels=2:8)
ggplotly(pmc2)
```



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Adopting a fully probabilistic modelling approach gives more satisfying results than the ‘fit statistics’ we had last week

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Example: penguins (2/2)

- Code

Best model: VEE, 4

What's wrong with this fit?

model: VEE, 3

Which is the better model? 4 or 3 clusters? Purely based on how well it fits the data?

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Summary

- Model-based clustering provides a nice automated clustering, if the data has neatly separated clusters, even in the presence of nuisance variables.
- Non-elliptical clusters could be modeled by combining multiple ellipses.
- It is affected by nuisance observations, and has a parameter `noise` to attempt to filter these.
- It may not function so well if the data hasn't got separated clusters.
- k-means and Wards linkage hierarchical would yield similar results to constraining the variance-covariance model to EEI (or VII, EEE).
- Having a functional model for the clusters is useful.

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Self-organising maps

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Overview

- A self-organizing map is a constrained k-means algorithm
- A 1D or 2D net is stretched through the data. The knots in the net form the cluster means, and points closest to the knot are considered to belong to that cluster.
- The net provides a low-dimensional summary of the clustering, nodes (and their corresponding clusters) that are close to each other being more similar than those that are further apart.

Source: wikipedia

The Algorithm

1. Scale your data
2. Initialize the net defined by the knots (nodes $m_k, k = 1, \ldots, K$): choose the number of nodes in the horizontal (and vertical directions for 2D), and set initial positions of these K nodes (eg first two PCs) in the data space.
3. Loop over data points, $x_i, i = 1, \ldots, n$
   i. find the closest node, m_{k^*}
   ii. for each node, m_k in the neighborhood of m_{k^*} and update it by: $m_k = m_k + \alpha h_k(x_i, m_{k^*}) \cdot (x_i - m_{k^*})$, pulling it closer to x_i,

   where α is a learning rate function that linearly shrinks from 1 to 0, or function with decreasing value as the number of iterations increases, and h_k is a neighbourhood function, e.g. $h_k(x_i, m_{k^*}) = \exp(\frac{||x_i - m_{k^*}||}{2\alpha})^2)$ which is a bubble function (within a distance or not).

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Step 3 is iterated until nodes stop changing position or a stopping rule is satisfied.

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Example: penguins (1/2)

kohonen

somInit

som

```
set.seed(947)
p_grid <- kohonen::somgrid(xdim = 5, ydim = 5,
                           topo = 'rectangular')
p_init <- somInit(as.matrix(p_std[,2:5]), 5, 5)
p_som <- som(as.matrix(p_std[,2:5]),
             rlen=2000,
             grid = p_grid,
             init = p_init)
```

`rlen` controls the length of the optimisation. Tend to need to run it for longer than default.

```
plot(p_som, type="changes")
```

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Example: penguins (2/2)

From the map we can see that the clustering has effectively distinguished the species, with some confusion between Chinstrap and Adelie.

The net is stretched and clumped into the three clusters in 4D.

Example: surface (1/3)

```
library(detourr)
set.seed(645)
detour(c3,
       tour_aes(projection = 1:10)) |>
       tour_path(grand_tour(2), fps = 60,
                 max_bases=40) |>
       show_scatter(alpha = 0.7,
                    axes = TRUE)
```

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Example: surface (2/3)

```
c3_som <- som(as.matrix(c3),
  grid = kohonen::somgrid(5, 5, "hexagonal"))
```

```
plot(c3_som, type="changes")
```

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Example: surface (3/3)

- the net stretches into the vertices of the tetrahedron, filling the smaller tetrahedrons
- see the break so that a 2D net fits the 3D object?
- the 7 noise dimensions were ignored

Next: Evaluating your clustering model

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